Machine Learning Analysis of Password Strength

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Introduction

As text passwords remain the default for password authentication, there is a growing need for public understanding of password strength, and what features reliably quantify it. Existing researchers rely on the LUDS mechanism: lowercase, uppercase, digits, and symbols to rank the strength of a password, though prior studies conclude that these quantitative heuristics are ineffective and often overlook linguistic patterns and other common word structures (ex. l33tspeak).

To address these disparities, researchers suggest using entropy-based metrics, such as zxcvbn, an open-source library. In this work, we offer a bridge between both approaches and present a Machine Learning (ML) based framework to evaluate password strength by fine-tuning a BERT model (bert-base-cased) on a labeled dataset of 468k passwords.

Objectives

- Analyze structural and linguistic patterns that contribute to password bias and vulnerability.
- Fine-tune a BERT-based classifier to learn contextual dependencies beyond rule-based heuristics.
- Evaluate relationships between password features (LUDS) and strength classifications.
- Apply graphical modeling and clustering to identify structural patterns.
- Compare learned models with existing strength estimators (e.g., zxcvbn).

Dataset & Tools

Kaggle Dataset

- Labeled password dataset (468,747 entries).
- Randomly generated password dataset (10,000 passwords).

Tools & Libraries

- BERT (bert-base-cased), HuggingFace Transformers
- PyTorch
- Pandas, NumPy
- Scikit-learn
- Matplotlib

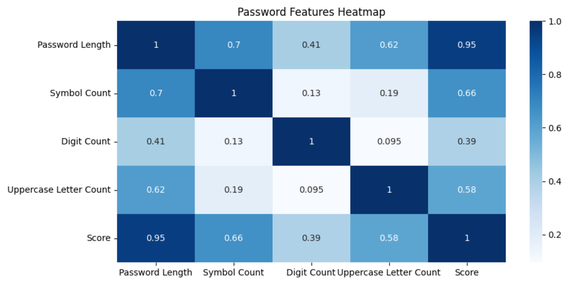
Methods

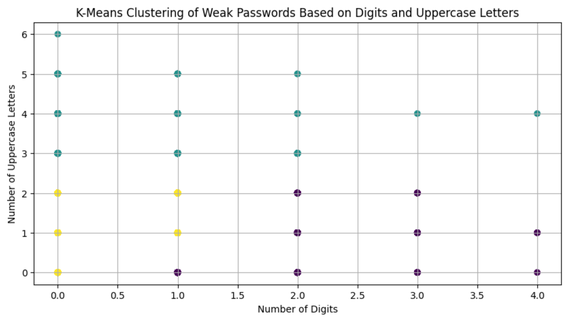
1. **Dataset:** The labeled Kaggle dataset was split into 70% training and 30% testing. After training, the model was applied to unlabeled random-password datasets to analyze learned patterns, correlations, and clustering behavior.
2. **Model:** Fine-tuned bert-base-cased with a linear classifier (max length 512), trained for 3 epochs using AdamW and cross-entropy loss.
3. **Evaluation:** Achieved 87% accuracy and 0.81 F1 score. Evaluated using correlation analysis and KMeans clustering.

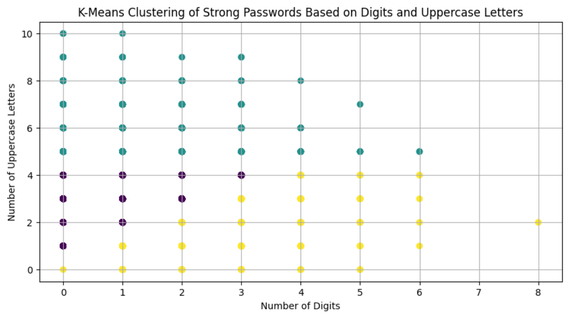
```
Training Loss: 0.3506726136717016
Arruracy: 0.8690702222222222
F1 Score: 0.8133530671739548
```

Results

- Password length (0.85) and symbol usage (0.57) showed the strongest correlation with strength.
- KMeans clustering showed weak passwords were homogeneous, while strong passwords were more diverse.
- BERT and zxcvbn showed high agreement, with differences in medium-strength rankings.

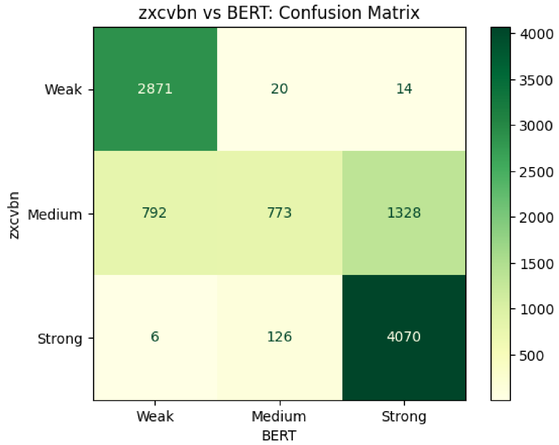






BERT vs Zxcvbn

BERT and zxcvbn shared a general consensus for most passwords. Differences lied in medium-strength passwords, as zxcvbn relies on pattern-matching rules while BERT evaluates linguistic features.



Conclusion

Overall, our results show that there is a correlation between password length, uppercase characters, symbols, and digits, and overall password strength. These features are also classified within LUDS, which suggests that there is a connection between BERT's classifications and the theory behind LUDS. Since there is also agreement between BERT and zxcvbn, we can deduce that zxcvbn still draws upon some of the same principles. While BERT and zxcvbn differ in how they classify medium-strength passwords, both approaches rely on structural characteristics of passwords when determining strength. This suggests that learned models such as BERT can complement existing heuristic-based approaches by capturing patterns that extend beyond fixed rules.